

# **Agentic Graph Reasoning**

## **Autonomous Knowledge Graph Navigation for Fact Verification and Hallucination Mitigation in Large Language Models**

Md. Sakif Khan    0421052099

Department of Computer Science and Engineering  
Bangladesh University of Engineering and Technology

Supervisor: Dr. Sadia Sharmin, Associate Professor

Pre-defense

# The problem

Language models answer factual questions **fluently**, and just as fluently when they do not hold the fact.

In our own no-retrieval control, on WebQSP:

- 661 entities asserted
- 179 of them **do not exist** in the knowledge graph
- that is 27.1%, stated with no hedge

## The gap

Fluency and factuality are produced by the same mechanism, so the output gives the reader no signal about which one they are getting.

A system that is confidently wrong is worse than one that says it does not know.

**The failure is not rare, and it is not visible in the answer.**

## Where existing approaches stop

| Approach           | What it does                        | Where it stops                                   |
|--------------------|-------------------------------------|--------------------------------------------------|
| Parametric         | Answers from weights                | No external evidence at all                      |
| Vector RAG         | Retrieves text once, then reasons   | Retrieval happens <i>before</i> reasoning begins |
| Static GraphRAG    | Retrieves a fixed neighbourhood     | Radius fixed before the question is understood   |
| Agentic navigation | Interleaves retrieval and reasoning | Emits whatever the last generation call produces |

**Nobody checks what the answer asserts *before* it is delivered.**

# Research questions

- RQ1** Does agentic navigation improve multi-hop factual accuracy — and does the advantage **grow with hop count**?
- RQ2** Given a system that already navigates the graph, what does a claim-level check against traversed triples **contribute**?
- RQ3** Which components earn their cost, in accuracy and in **tokens**?

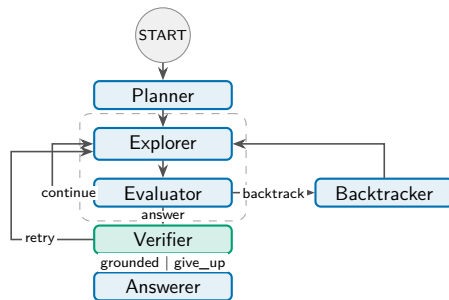
RQ2 is posed as *what does it contribute*, not *does it reduce hallucination*. The answer has several parts, and a yes/no framing would hide them.

# AGR: an explicit state machine

Six nodes; the model is consulted only at decision points.

- **Planner** — sub-objectives
- **Explorer** — expands the frontier
- **Evaluator** — objective met?
- **Backtracker** — undo + ban list
- **Verifier** — the contribution
- **Answerer** — emits what survived

Two cycles, both bounded by budgets rather than by model behaviour.



## Constrained tools, not free-form queries

| Operation                      | Returns                         | Why it is capped                 |
|--------------------------------|---------------------------------|----------------------------------|
| <code>get_relations</code>     | Candidate relations             | $\leq 300$ offered to the scorer |
| <code>get_neighbors</code>     | Neighbours of an entity         | $\leq 200$ per expansion         |
| <code>verify_connection</code> | Adjacency in the full graph     | Used only by the verifier        |
| <code>link_entity</code>       | Surface form $\rightarrow$ node | Three-stage resolver             |

The agent never writes Cypher; every operation is logged.

**Determinism in the tools is what makes the verification layer possible.**

# The Structural Verification Layer

Before anything is emitted:

1. Draft answer is split into **atomic claims**
2. Each claim is checked against the **triples actually traversed**
3. Supported claims keep their supporting triples
4. Unsupported claims trigger **targeted re-exploration**, or are **dropped**

The system therefore **hedges rather than asserts**, and returns every answer with its evidence.

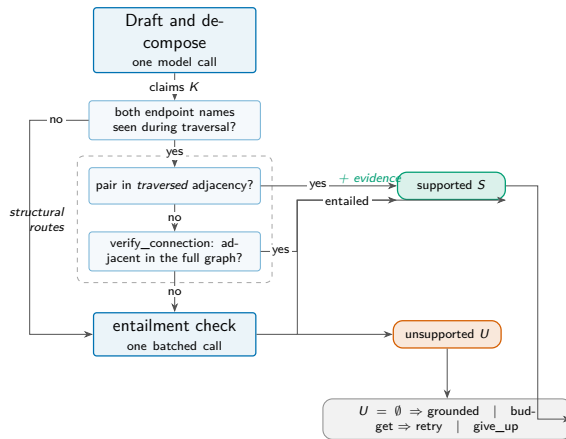
## *Why structural*

The check is against the graph the agent walked — not against the model's own judgement of its own output. A model asked to grade itself is not an independent check.

## *The honest limit*

A claim can be *true* and still be the wrong answer. Structural grounding cannot catch that — see the echo attractor later.

# One claim, three routes



Only the third route costs a model call. The first two are pure graph lookups against the traversal record.



# The environment and the question sets

## Freebase-derived graph

|                    |           |
|--------------------|-----------|
| Entities           | 2,592,892 |
| Triples            | 8,309,194 |
| Distinct relations | 7,058     |
| Import time        | 36.4 s    |

## Certified question sets

|               | WebQSP | CWQ   |
|---------------|--------|-------|
| Questions     | 400    | 400   |
| Gold (median) | 1.5    | 1.0   |
| Reachable     | 97.0%  | 99.2% |
| Multi-hop     | 132    | 260   |

Reachability is the *ceiling* on every Hits@1 reported: a miss is the system's, not the environment's.

**800 questions, with reachability certified per question so the accuracy ceiling is known.**

# Making the comparison fair

Five systems, **one frozen backbone, one budget:**

- No-retrieval (parametric control)
- Vector RAG
- Static GraphRAG
- Think-on-Graph (agentic)
- **AGR**

Same model, temperature 0, same 25-call cap, same questions, same graph.

## What this buys

Differences are attributable to **architecture**, not to model capacity or to a bigger retrieval budget.

## Why a parametric control

These benchmarks predate current models and may be partly memorised. Without the control we could not separate retrieval from recall.

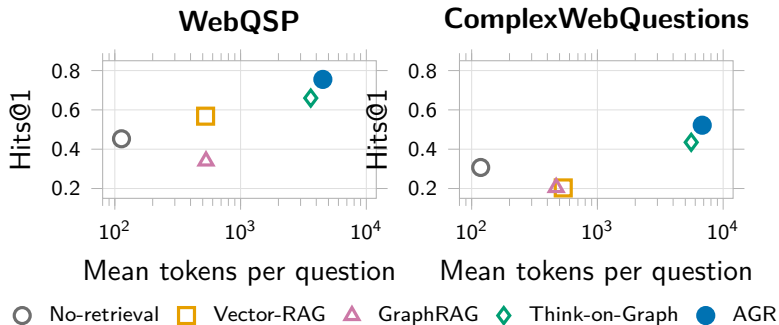
## Main results

| System          | WebQSP       |              | ComplexWebQuestions |              | Cost / question |            |
|-----------------|--------------|--------------|---------------------|--------------|-----------------|------------|
|                 | Hits@1       | F1           | Hits@1              | F1           | Tokens          | Calls      |
| No-retrieval    | 0.453        | 0.305        | 0.307               | 0.279        | 113             | 1.0        |
| Vector RAG      | 0.568        | 0.432        | 0.203               | 0.168        | 527             | 1.0        |
| Static GraphRAG | 0.340        | 0.262        | 0.205               | 0.183        | 531             | 1.0        |
| Think-on-Graph  | 0.660        | 0.477        | 0.435               | 0.378        | 3,615           | 12.8       |
| <b>AGR</b>      | <b>0.755</b> | <b>0.642</b> | <b>0.522</b>        | <b>0.469</b> | 4,511           | <b>6.2</b> |

Cost is WebQSP. On CWQ: AGR 6,818 tokens and 8.9 calls, Think-on-Graph 5,572 and 18.2.

**Ahead of every baseline on both datasets, at half the model calls of the other agentic system.**

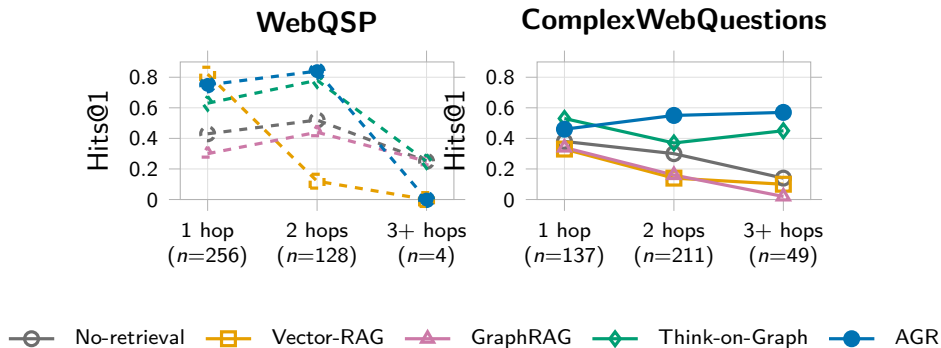
# Accuracy against cost



On *tokens* Think-on-Graph is **not** dominated — it buys lower accuracy at a lower token price. On *calls*, which the budget meters, it is.

**The frontier depends on which cost you charge.**

## RQ1: accuracy against hop count



On CWQ, AGR goes  $0.46 \rightarrow 0.55 \rightarrow 0.57$  — the **only** system that ends above where it started. Three of the other four decay monotonically.

RQ1 is answered by a shape, not a difference of means.

## The caveat I want to raise myself

| Hits@1 on...                       | WebQSP          |              | ComplexWebQuestions |              |
|------------------------------------|-----------------|--------------|---------------------|--------------|
|                                    | ToG             | AGR          | ToG                 | AGR          |
| questions ToG <i>finishes</i>      | <b>0.852</b>    | 0.788        | <b>0.629</b>        | 0.607        |
| questions ToG is <i>clipped</i> on | 0.197           | <b>0.675</b> | 0.188               | <b>0.415</b> |
| clip rate                          | 29.2% (117/400) |              | 44.0% (176/400)     |              |

Where Think-on-Graph finishes inside the shared 25-call cap, it is **ahead of AGR on both datasets**. The entire aggregate margin comes from the questions it cannot finish.

**The claim is efficiency under a fixed budget — not that AGR reasons better per step.**

## RQ2: does anything ungrounded get asserted?

| System          | Entities asserted | Ungrounded | Rate        |
|-----------------|-------------------|------------|-------------|
| No-retrieval    | 1,001             | 221        | 22.1%       |
| Vector RAG      | 1,329             | 45         | 3.4%        |
| Static GraphRAG | 1,024             | 8          | 0.8%        |
| Think-on-Graph  | 1,265             | 0          | <b>0.0%</b> |
| <b>AGR</b>      | 1,709             | 0          | <b>0.0%</b> |

Both datasets pooled. AGR asserts no entity that does not exist in the graph, across 1,709 assertions.

**But Think-on-Graph reaches 0.0% too** — so this is a property of *graph navigation*, not of the verification layer.

## RQ2: so what does verification actually contribute?

### What it does *not* do

Removing it changes accuracy by an amount we cannot detect:  $p = 1.0$  on *both* datasets. It does not earn its place on Hits@1 or F1.

### What it does do

- Turns silent error into a **hedge**: AGR hedges on 8.2% of WebQSP against no-retrieval's 12.2% error rate
- Attaches **supporting triples** to every asserted claim
- Gives an **auditable output contract**

This is a negative result reported as one. The layer's case rests on auditability, and the thesis says so rather than claiming an accuracy benefit the measurement does not support.



## RQ3: what each component earns

| Condition              | WebQSP       |              | ComplexWebQuestions |       | Tokens (WQ)  |
|------------------------|--------------|--------------|---------------------|-------|--------------|
|                        | F1           | $p$          | F1                  | $p$   |              |
| Full system            | 0.659        | —            | 0.445               | —     | 4,562        |
| No planner             | <b>0.742</b> | <b>0.006</b> | 0.398               | 0.088 | <b>3,159</b> |
| No backtracking        | 0.646        | 0.727        | 0.445               | 1.000 | 4,261        |
| No verifier            | 0.663        | 1.000        | 0.436               | 1.000 | 4,460        |
| Embedding-only scoring | 0.643        | 0.664        | 0.437               | 0.481 | 3,413        |

Paired McNemar against the full system, stratified halves ( $n = 200 / 198$ ).

**Exactly one component shows a detectable accuracy effect — and the sign is backwards.**

# The result I did not expect

## Removing the planner *improves* WebQSP:

- +0.083 F1, at  $p = 0.006$
- -31% tokens
- 6.2  $\rightarrow$  4.0 calls

On ComplexWebQuestions it trends the other way ( $-0.047$  F1,  $p = 0.088$ ) — not significant, but the opposite sign.

## Why

WebQSP is predominantly one hop. Decomposing a one-hop question invents a second sub-objective that sends the frontier away from the answer.

## What follows

Decomposition should be **gated on question structure**, not applied unconditionally. That is a design conclusion the measurement forced, not one it confirmed.

## Every failure, read: the echo attractor

All 259 remaining failures read and labelled, both datasets pooled:

| Category            | n  |
|---------------------|----|
| Relation selection  | 65 |
| Composite claim     | 47 |
| Knowledge-graph gap | 44 |
| Decomposition error | 38 |
| Answer selection    | 15 |
| Echo attractor      | 13 |

The characteristic error of a graph navigator is **not invention**. It is the **echo attractor**: a real, grounded entity one hop from the answer.

### Why it matters

Any grounding check *passes* it — because it is true. Verification cannot catch it by construction.

It appears across systems, so it is a property of the task, not of AGR. Naming it is what lets future work target it.

# The benchmark was wrong 57 times

Reading every failure also found questions where the *benchmark*, not the system, was at fault:

- 41 excluded before the census
- 17 found inside the census
- 1 counted in both

⇒ 57 **distinct questions**

**A contribution that outlives this particular system.**

## Why report it

Reported defect rates for these two standard benchmarks are rare and usually anecdotal. This is a counted rate with a documented adjudication procedure and per-question provenance.

# Contributions, and what I would not claim

## Contributions

1. AGR: a state machine over a constrained graph tool API
2. The Structural Verification Layer and its output contract
3. A controlled five-system comparison, one backbone, one budget
4. The hop-count shape answering RQ1
5. The *echo attractor*, named
6. A counted benchmark-defect rate (57)

## Limitations I state plainly

- Verification earns **no** detectable accuracy gain
- Zero ungrounded assertion is **navigation**, not the layer
- ToG leads on the questions it finishes
- One backbone: results are architectural, not universal

**Thank you**

Questions